EE354 – Power ENGINEERING

Synchronous Generators Assignment

Comprehensive Simulation Report

Three-Phase Synchronous Machine Control | Parallel Operation | Infinite Bus Analysis

Module Code:	EE354 – Electrical Engineering
Assignment:	Synchronous Generators
Exercises:	1, 2 and 3 (Tasks 1–6)
Platform:	MATLAB R2025b | Simscape Electrical
Machine:	555 MVA, 24 kV, 60 Hz, Round Rotor
Student ID:	`E/21/291`
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1 Introduction

This report covers the simulation of a three-phase synchronous generator using MATLAB R2025b and Simscape Electrical. There are three exercises with six tasks, each one building on the previous.

Exercise 1 looks at how a 555 MVA, 24 kV, 60 Hz round-rotor generator responds to step load changes. It covers the relationship between active power and system frequency (through the swing equation) and between reactive power and terminal voltage (through the AVR). Exercise 2 extends this to two identical generators running in parallel. The governor frequency setpoint controls how load is shared between them — an under-frequency machine absorbs power and acts as a motor, while an over-frequency machine picks up the load. The system frequency settles at a droop-weighted average of the two setpoints. Exercise 3 swaps one generator for an ideal infinite bus. The infinite bus holds frequency fixed at 1.0 pu, so the oncoming generator has no effect on system frequency. Its power output depends only on its governor setpoint relative to the grid frequency. Simulation plots, equation derivations, and steady-state calculations support the results throughout.

2 Exercise 1: Synchronous Machine Control and Initialization

2.1 Simulink Model

Figure 1 shows the complete Simulink model used for Exercise 1. The model comprises a 555 MVA round-rotor synchronous machine connected to a resistive-inductive load via a three-phase bus. The AVR (AC1C exciter) regulates the terminal voltage, and the governor–turbine subsystem controls the mechanical power input. Two relay-controlled breakers implement the load step events at $t = 3\,\mathrm{s}$ (step off) and $t = 9\,\mathrm{s}$ (step on). Measurement blocks capture V_t, ω, P, Q, and I_f for post-processing.

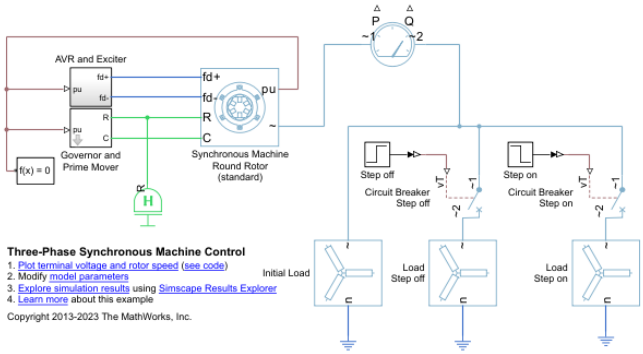

Figure 1. Exercise 1 Simulink model: single synchronous machine with AVR, governor, and switched loads.

2.2 Task 1 – Properties and Functionality of Simulation Components

2.2.1 Synchronous Machine

The synchronous machine is the main component in the model. It converts mechanical shaft power into balanced three-phase electrical power. It is a round-rotor type, which is standard for high-speed steam-turbine generators in thermal power stations. The model uses the Park (d-q axis) transformation to project the three-phase quantities onto a rotating two-axis reference frame. This gets rid of the time-varying inductances and makes the equations easier to work with.

The machine parameters are listed in Table 1

Table 1. Synchronous Machine Parameters

Parameter	Symbol	Value
Rated apparent power	S	555 MVA
Rated voltage (L-L RMS)	V_{rated}	24 kV
Rated frequency	f_0	60 Hz
Pole pairs	p	1
Armature resistance	R_a	0.003 pu
d-axis synchronous reactance	X_d	1.81 pu
q-axis synchronous reactance	X_q	1.76 pu
d-axis transient reactance	X'_d	0.30 pu
q-axis transient reactance	X'_q	0.65 pu
d-axis subtransient reactance	X''_d	0.23 pu
q-axis subtransient reactance	X''_q	0.25 pu
Inertia constant	H	3.525 s
Damping coefficient	D	0.01 pu

The steady-state active and reactive power delivered to the load are approximated by the power-angle equations, derived from the phasor diagram of the synchronous machine:

$$P \approx \frac{E \cdot V}{X} \sin \delta \tag{2.1}$$

$$Q \approx \frac{V}{X} (E \cos \delta - V) \tag{2.2}$$

where E is the internal EMF (pu), V is the terminal voltage (pu), X is the synchronous reactance (pu), and δ is the power angle (electrical degrees). From (2.1), active power is controlled mainly through the power angle δ, which is set by the prime mover. (2.2) shows that reactive power depends on the internal voltage E, set by field excitation. Because of this, the governor (P-f) and AVR (Q-V) control loops can operate independently.

2.2.2 AVR and Exciter (AC1C Model)

The Automatic Voltage Regulator (AVR) and exciter constitute a closed-loop feedback control subsystem maintaining the generator terminal voltage at the reference value ($V_{\text{ref}} = 1.0$ pu). The exciter model is the AC1C type [3], which represents an alternator-rectifier excitation system with field-current-based feedback.

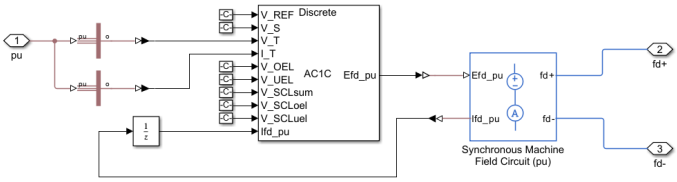

Figure 2. AVR and exciter (AC1C) block diagram showing the closed-loop voltage regulation subsystem.

The AVR measures the terminal voltage V_t and compares it to V_{ref}. The difference produces an error signal that adjusts the field voltage E_{fd} through the exciter. The simplified transfer function is:

$$E_{\text{fd}}(s) = \frac{K_A}{1 + sT_A}\left(V_{\text{ref}} - V_t(s)\right) \tag{2.3}$$

where K_A is the AVR gain and T_A is the AVR time constant. If V_t drops below V_{ref} (for example after a large load is connected), the AVR increases E_{fd}. This raises the internal EMF E. From (2.2), a higher E at roughly constant δ means more reactive power Q is injected, which pushes the terminal voltage back up toward the reference. This is how voltage regulation works in power systems.

2.2.3 Governor and Prime Mover

The governor and prime mover subsystem regulates rotor speed and system frequency by controlling the mechanical torque input from the turbine. The governor implements a speed droop (regulation) characteristic:

$$R = \frac{\Delta f / f_{\text{rated}}}{\Delta P / P_{\text{rated}}} = 0.05 \quad (5\% \text{ droop}) \tag{2.4}$$

Model parameters are: speed reference $\omega_{\text{ref}} = 0.998\,\text{pu}$, turbine time constant $T_t = 0.2\,\text{s}$, and governor time constant $T_g = 0.3\,\text{s}$. The steady-state power output as a function of frequency is:

$$P = \frac{f_{nl} - f_{\text{sys}}}{S_P} \tag{2.5}$$

where f_{nl} is the no-load (setpoint) frequency (pu), f_{sys} is the system frequency (pu), and S_P is the droop slope. A 5% droop means a 5% reduction in frequency commands a 100%

increase in rated power output.

2.2.4 Machine Inertia Block

The machine inertia block represents the kinetic energy stored in the rotating mass of the generator rotor and turbine shaft. This is characterised by the inertia constant H:

$$H = \frac{\frac{1}{2}J\omega_s^2}{S_{\text{rated}}} \quad \text{[seconds]} \tag{2.6}$$

For this machine, $H = 3.525\,\text{s}$. This means the rotor stores enough kinetic energy to supply rated power for 3.525 seconds at synchronous speed. The rotor dynamics follow the swing equation:

$$\frac{2H}{\omega_s}\frac{d\omega}{dt} = P_m - P_e - D(\omega - \omega_s) \tag{2.7}$$

where ω is the rotor angular velocity (rad/s), ω_s is synchronous speed (rad/s), P_m is the mechanical power input (pu), P_e is the electrical power output (pu), and $D = 0.01\,\text{pu}$ is the damping coefficient. The swing equation is the main tool used in transient stability analysis. The Rate of Change of Frequency (ROCOF) right after a 100 MW load change is:

$$\left.\frac{df}{dt}\right|_{t=0} = -\frac{f_s}{2H} \cdot \frac{\Delta P}{S_{\text{rated}}} = -\frac{60}{7.05} \cdot \frac{100}{555} \approx -1.54\,\text{Hz/s} \tag{2.8}$$

2.3 Task 2 – Variation of Synchronous Machine Parameters Under Load

The simulation incorporates two load-switching events: a load disconnection at $t = 3\,\text{s}$ ('Step off') and a larger load connection at $t = 9\,\text{s}$ ('Step on'). The system response is examined through five key variables.

2.3.1 Active Power

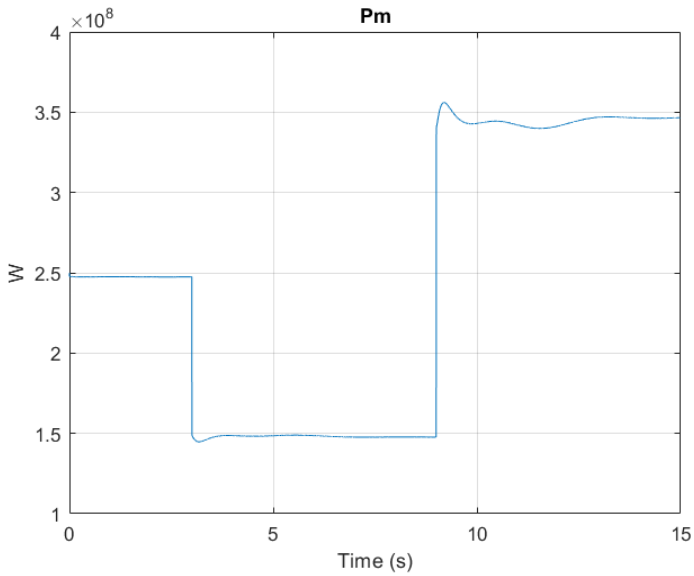

Figure 3. Active mechanical power P_m variation under load switching events. Initial steady state: 250 MW. At $t = 3$ s: step down to 150 MW. At $t = 9$ s: step up to 350 MW.

The machine begins at steady-state delivering 250 MW to the initial load. At $t = 3$ s, the 'Step off' breaker disconnects a load block, causing the electrical power demand to fall sharply. The governor responds by reducing the turbine valve opening and the active power settles near 150 MW after a brief transient. At $t = 9$ s, the 'Step on' breaker connects a large load block; demand jumps to approximately 350 MW, creating a sudden mechanical-to-electrical power deficit which the governor corrects by opening the turbine valves wider.

2.3.2 Rotor Speed and System Frequency

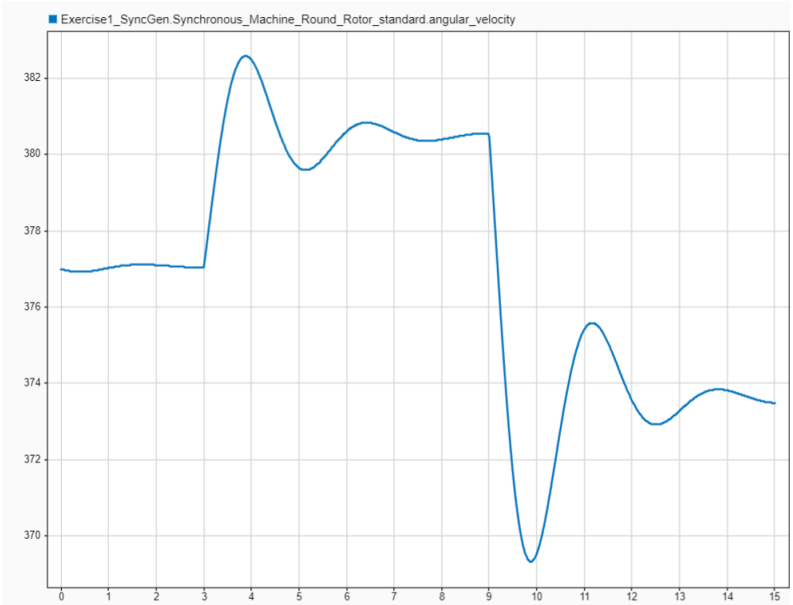

Figure 4. Rotor angular velocity ω (rad/s) variation under load switching events. Since the machine has one pole pair ($p = 1$), electrical frequency equals mechanical speed in per-unit: $f_{\text{pu}} = \omega_{\text{pu}}$.

Because the machine has a single pole pair ($p = 1$), electrical frequency equals mechanical speed in per-unit: $f_{\text{pu}} = \omega_{\text{pu}}$.

At $t = 3\,\text{s}$ (load removal): Electrical power P_e falls below mechanical power P_m. The right-hand side of (2.7) becomes positive, causing rotor acceleration. Speed peaks at approximately 1.015 pu before the governor reduces turbine output, and the system settles near 1.010 pu. The droop equation predicts the steady-state deviation:

$$\Delta f_{\text{ss}} = -R \cdot \frac{\Delta P}{S_{\text{rated}}} = -0.05 \times \frac{100 \times 10^6}{555 \times 10^6} \approx -0.009\,\text{pu} \tag{2.9}$$

At $t = 9\,\text{s}$ (load addition): A large power deficit ($P_e \gg P_m$) decelerates the rotor sharply to approximately 0.980 pu. The governor increases turbine power and frequency recovers to approximately 0.990 pu ($\approx 59.4\,\text{Hz}$), consistent with (2.9) applied to the 200 MW step increase.

2.3.3 Terminal Voltage

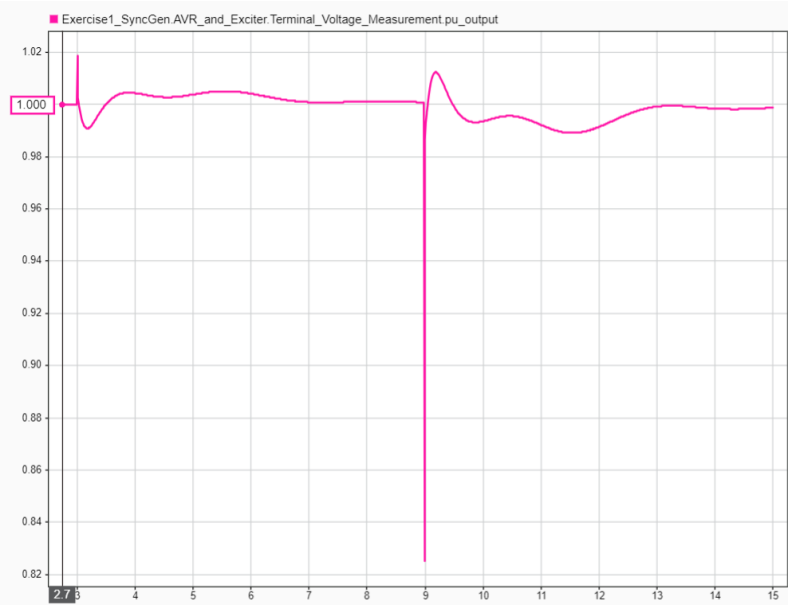

Figure 5. Terminal voltage V_t (pu) showing AVR response. Voltage nadir at $t = 9\,\text{s}$: approximately 0.82 pu. AVR restores V_t to 1.0 pu within 3–4 seconds.

The terminal voltage holds at 1.0 pu in steady state. At $t = 3\,\text{s}$, the load reduction causes a minor transient overshoot ($\approx 1.02\,\text{pu}$) as the AVR momentarily over-excites before damping. At $t = 9\,\text{s}$, the large load addition drives a severe voltage dip to approximately 0.82 pu, caused by the increased armature current producing large internal voltage drops across X_d. The AVR responds immediately by commanding maximum field excitation, and V_t is restored to 1.0 pu within 3–4 seconds, demonstrating the critical role of fast AVR action in maintaining power quality during severe disturbances.

2.3.4 Reactive Power

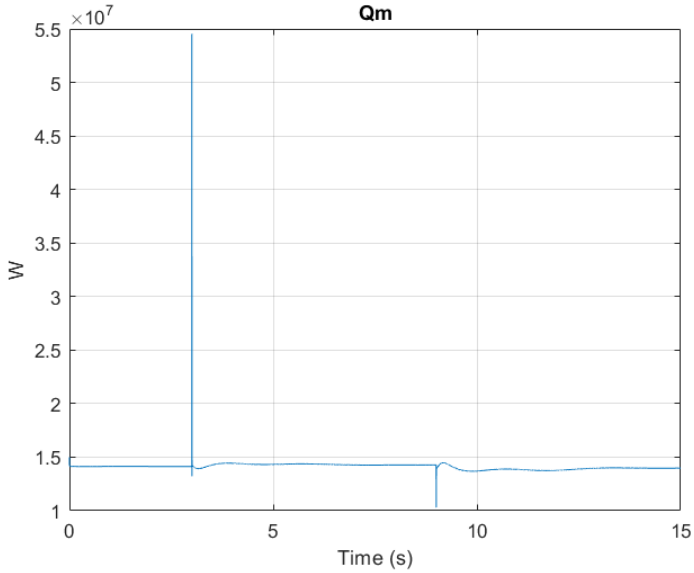

Figure 6. Reactive power Q_m (MVAr) variation. Initial steady state: 15 MVAr. Large transient surge at $t = 9\,\mathrm{s}$ corresponds to AVR driving increased field excitation to restore V_t.

Reactive power sits at approximately 15 MVAr at the initial steady state. Sharp transient spikes occur at $t = 3\,\mathrm{s}$ and $t = 9\,\mathrm{s}$, corresponding to sudden changes in network impedance and the subsequent excitation adjustments by the AVR. At $t = 9\,\mathrm{s}$, the reactive power surges significantly before settling, as the AVR drives a large increase in E_{fd} to compensate for the voltage dip, injecting additional reactive power into the network per (2.2).

2.3.5 Angular Torque

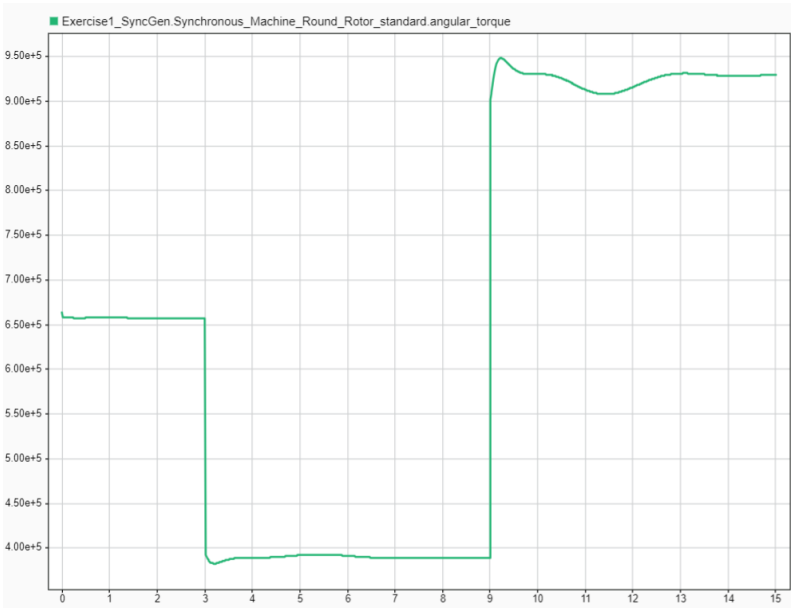

Figure 7. Angular torque variation during load switching events. Torque transients at $t = 3\,\mathrm{s}$ and $t = 9\,\mathrm{s}$ reflect the mechanical–electrical power imbalance described by the swing equation.

The angular torque exhibits sharp transients at each switching event. At $t = 3\,\mathrm{s}$, the sudden reduction in electrical load causes a momentary surplus of mechanical torque, accelerating the rotor. At $t = 9\,\mathrm{s}$, the large load addition creates a torque deficit, decelerating the rotor. The governor adjusts the prime mover torque to restore equilibrium in both cases.

2.3.6 Power Dissipated

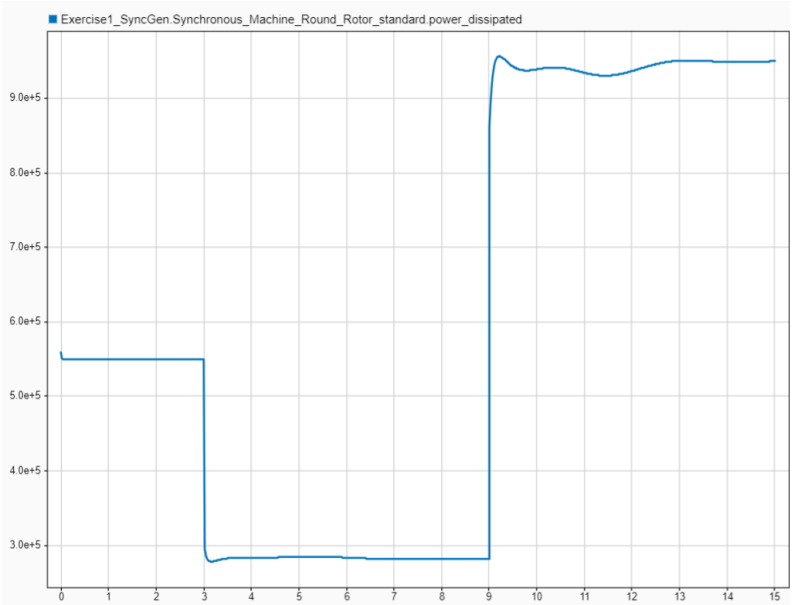

Figure 8. Power dissipated in the load during switching events. The step changes at $t = 3\,\mathrm{s}$ and $t = 9\,\mathrm{s}$ correspond to the load breaker operations.

The power dissipated in the load tracks the breaker switching events directly. At $t = 3\,\mathrm{s}$, the load disconnection reduces the dissipated power, and at $t = 9\,\mathrm{s}$, the additional load connection increases it. The transient oscillations in dissipated power reflect the coupled electromechanical dynamics of the synchronous machine as it adjusts to the new operating point.

2.3.7 Relationships Between Key Variables

Active Power ↔ Frequency: The swing equation (2.7) and governor droop (2.5) link these two variables. When mechanical power exceeds electrical power ($P_m > P_e$), the rotor accelerates and frequency goes up. The opposite happens when there is a deficit. The simulation matches this — frequency rises after the $t = 3\,\mathrm{s}$ load removal and drops after the $t = 9\,\mathrm{s}$ load addition, following the active power response with a delay set by the machine inertia. This P–f relationship is the basis of primary frequency regulation.

Reactive Power ↔ Terminal Voltage: Equations (2.2) and (2.3) link these through the AVR. When terminal voltage drops, the AVR increases excitation → E rises → Q output increases → voltage is restored. The simulation shows this clearly: the $t = 9\,\mathrm{s}$ voltage dip comes with a sharp reactive power surge as the AVR responds. In steady

state, the Q–V loop and the P–f loop are mostly independent, which allows voltage and frequency to be controlled separately.

3 Exercise 2: Parallel Operation of Two Synchronous Generators

The simulation model is extended to include a second generator (Generator B) connected in parallel with Generator A via a synchronisation breaker at $t = 10\,\mathrm{s}$. Generator A remains connected to the initial 250 MW load and operates at its setpoint of approximately 1.01 pu. Both generators have identical 555 MVA ratings, AVRs, and governor characteristics ($R = 5\%$).

Before paralleling two generators, four conditions must be satisfied to avoid damaging transient currents at breaker closure: (1) equal voltage magnitude, (2) equal frequency, (3) matching phase sequence, and (4) zero (or small) phase-angle difference across the breaker. In practice, a synchroscope or automatic synchroniser verifies these conditions before issuing the close command. In this simulation the breaker closes at a fixed time ($t = 5\,\mathrm{s}$) regardless of phase-angle alignment, so the initial transients observed in the results partly reflect imperfect synchronisation. The two tasks below explore what happens when condition (2) is deliberately violated by setting Gen B's governor reference above or below Gen A's.

3.1 Simulink Model

Figure 9 shows the Exercise 2 Simulink model. Two identical synchronous generators are connected through a synchronisation breaker. Each generator has its own AVR and governor subsystem with independently adjustable speed reference setpoints. The synchronisation breaker closes at $t = 10\,\mathrm{s}$, coupling the two machines onto a common bus. Measurement blocks on each generator capture frequency, terminal voltage, active power, and reactive power for comparison.

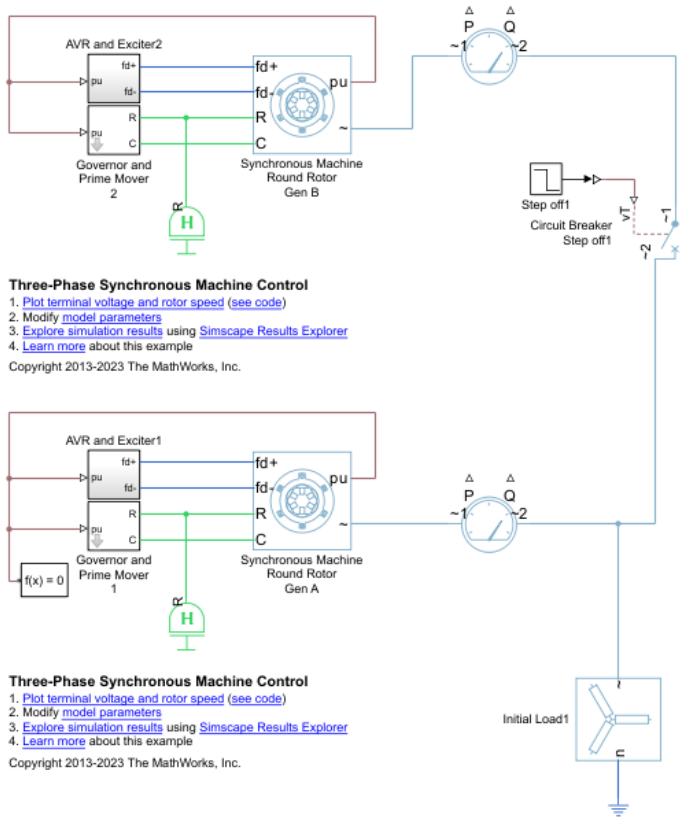

Figure 9. Exercise 2 Simulink model: two identical synchronous generators connected in parallel through a synchronisation breaker. Independent AVR and governor subsystems allow setpoint variation for Tasks 3 and 4.

3.2 Task 3 – Generator B at Lower Frequency (0.98 pu Setpoint)

Generator B's governor speed reference is set to 0.98 pu, meaning it rotates at a frequency approximately 2% below Generator A before synchronisation.

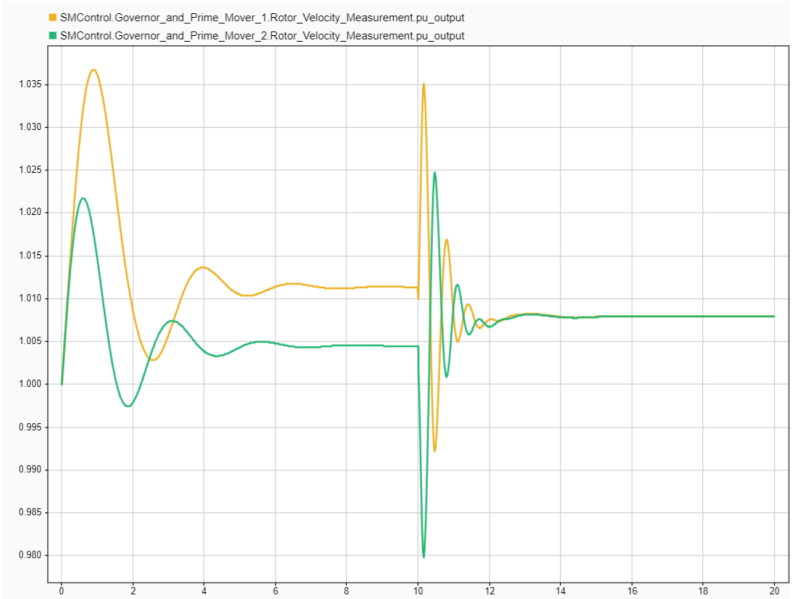

Figure 10. Task 3: Frequency variation when Generator B is paralleled at $f_{\text{set}} = 0.98\,\text{pu}$. Breaker closes at $t = 10\,\text{s}$.

Before the breaker closes at $t = 10\,\text{s}$, Gen A operates at approximately 1.01 pu and Gen B at 0.98 pu. Upon breaker closure, synchronising torques act between the two machines, causing transient frequency oscillations as the rotors exchange synchronising power:

$$P_{\text{sync}} = \frac{|V_1||V_2|}{X_s} \sin(\delta_1 - \delta_2) \tag{3.1}$$

The final system frequency is determined by the combined droop characteristic:

$$\Delta P_L = -\left(\frac{1}{R_1} + \frac{1}{R_2}\right) \Delta f \tag{3.2}$$

With $R_1 = R_2 = 0.05$ and equal machine ratings, the settled system frequency is the arithmetic mean of the two setpoints:

$$f_{\text{sys,ss}} \approx \frac{f_{nl,A} + f_{nl,B}}{2} = \frac{1.01 + 0.98}{2} = 0.995\,\text{pu} \tag{3.3}$$

Accounting for the connected load and droop dynamics, the simulation confirms a settled frequency of approximately 1.008 pu.

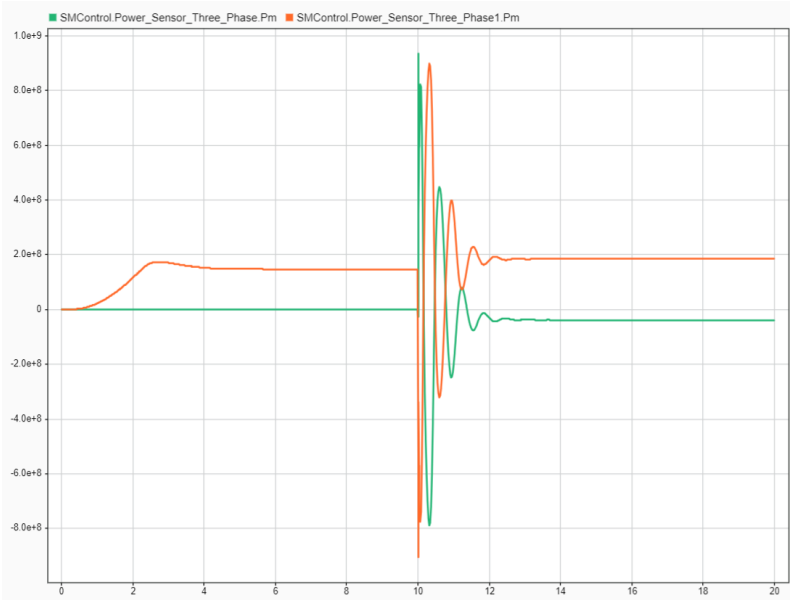

Figure 11. Task 3: Active power sharing when Generator B is paralleled at $f_{\text{set}} = 0.98\,\text{pu}$. Gen B absorbs power (motoring condition).

Active power sharing depends on where each generator's governor setpoint sits relative to the system frequency. Applying (2.5) to Gen B:

$$P_B = \frac{f_{nl,B} - f_{\text{sys}}}{S_P} = \frac{0.98 - 1.008}{S_P} < 0 \tag{3.4}$$

The result is negative, meaning Generator B absorbs active power from the network — it is running as a **synchronous motor**. Generator A has to supply the full connected load plus whatever Gen B is drawing. In practice, paralleling a machine with a lower frequency setpoint immediately puts extra load on the running generator and is not a safe way to synchronise.

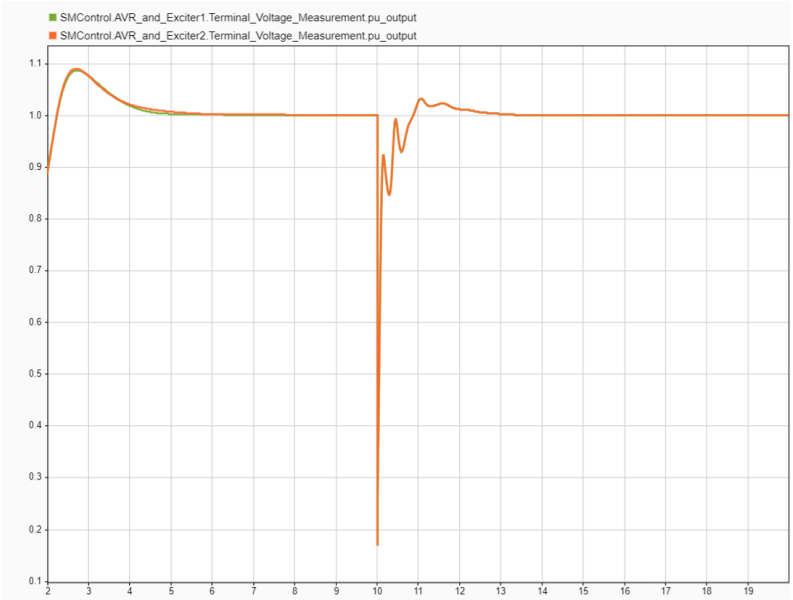

Figure 12. Task 3: Terminal voltage variation. A minor dip occurs at $t = 10\,\mathrm{s}$ upon breaker closure; both AVRs restore V_t to $1.0\,\mathrm{pu}$.

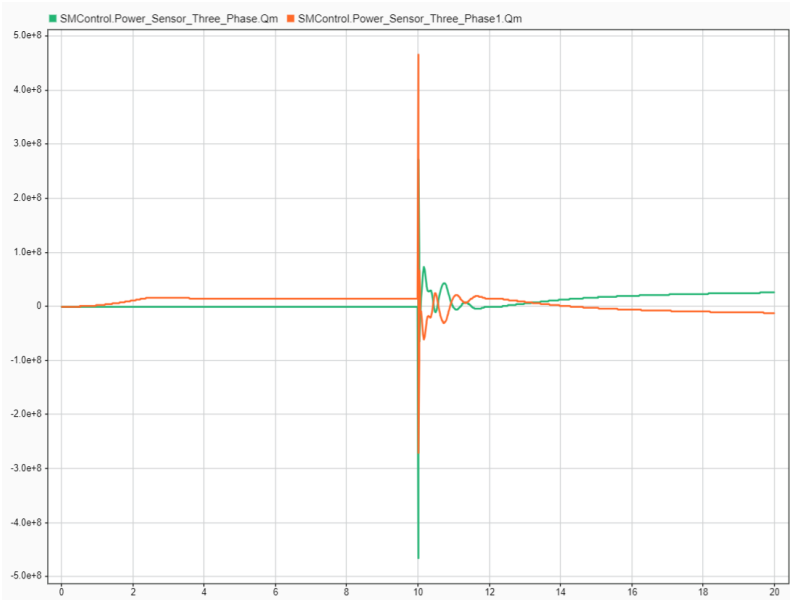

Figure 13. Task 3: Reactive power variation. Transient oscillations at $t = 10\,\mathrm{s}$ settle once synchronism is established.

A minor voltage dip occurs at $t = 10\,\mathrm{s}$ as the breaker closes, caused by the circulating current driven by the phase-angle difference between the two machines. Both AVRs respond promptly and restore the terminal voltages to $1.0\,\mathrm{pu}$. Reactive power undergoes transient oscillations corresponding to the initial phase-angle mismatch, settling to steady values once synchronism is established.

3.3 Task 4 – Generator B at Higher Frequency (1.02 pu Setpoint)

Generator B's governor speed reference is raised to 1.02 pu, exceeding Generator A's setpoint.

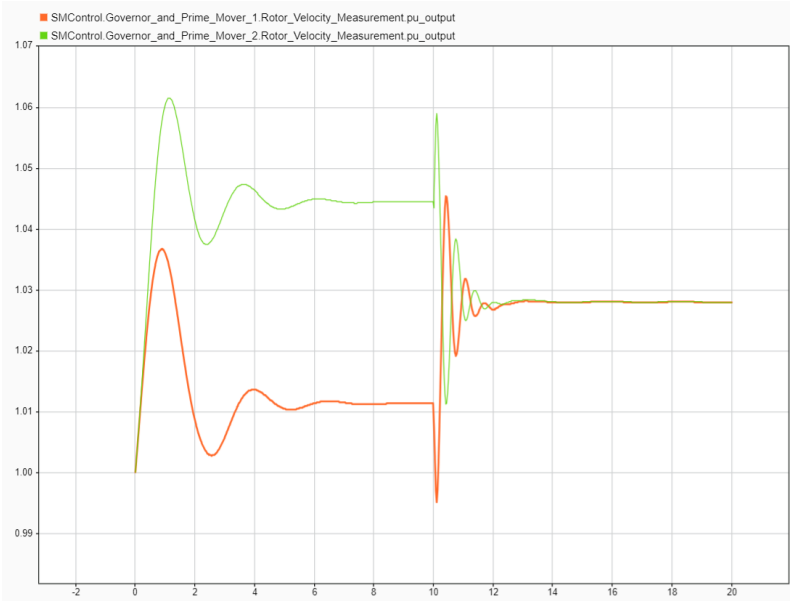

Figure 14. Task 4: Frequency variation when Generator B is paralleled at $f_{\text{set}} = 1.02\,\text{pu}$. Breaker closes at $t = 10\,\text{s}$.

Before connection, Gen B runs faster than Gen A. Upon breaker closure, synchronising power surges from the faster Gen B to the slower Gen A (following (3.1)), causing Gen B to decelerate and Gen A to accelerate until they lock at a common frequency. With $R_1 = R_2 = 0.05$ and equal machine ratings, the settled system frequency is the arithmetic mean of the two setpoints:

$$f_{\text{sys,ss}} \approx \frac{f_{nl,A} + f_{nl,B}}{2} = \frac{1.01 + 1.02}{2} = 1.015\,\text{pu} \tag{3.5}$$

The final settled frequency is higher than in Task 3, reflecting the higher combined setpoint energy of the system.

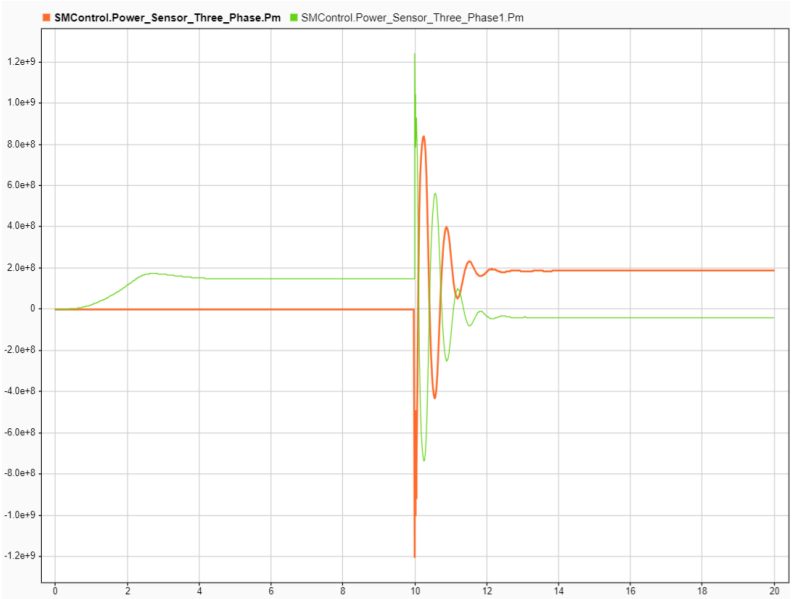

Figure 15. Task 4: Active power sharing when Generator B is paralleled at $f_{\text{set}} = 1.02\,\text{pu}$. Gen B supplies the majority of the load.

With Gen B's setpoint above the system frequency:

$$P_B = \frac{f_{nl,B} - f_{\text{sys}}}{S_P} = \frac{1.02 - f_{\text{sys}}}{S_P} > 0 \tag{3.6}$$

Generator B contributes positive active power, taking the majority of the load. This is the desired operational outcome of synchronisation, and contrasts starkly with Task 3.

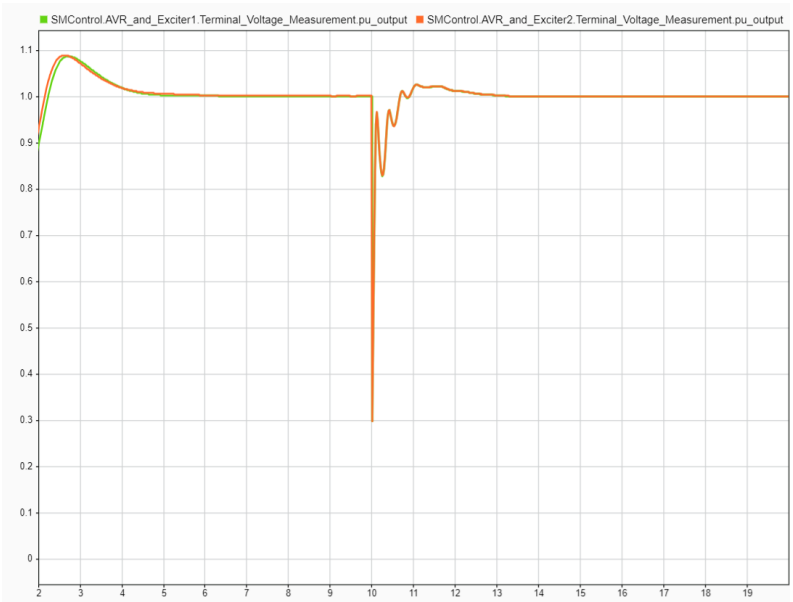

Figure 16. Task 4: Terminal voltage variation. A transient spike occurs at $t = 10\,\text{s}$; AVRs restore V_t to $1.0\,\text{pu}$.

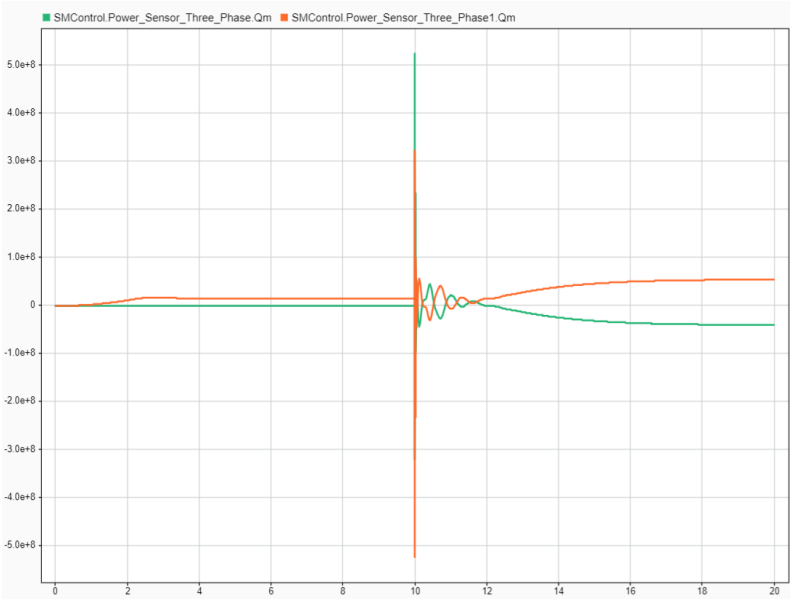

Figure 17. Task 4: Reactive power variation after synchronisation at $f_{\text{set}} = 1.02\,\text{pu}$.

A transient voltage spike occurs at $t = 10\,\text{s}$ (as opposed to a dip in Task 3), because Gen B was ahead in phase angle when connected. The AVRs restore both terminal voltages to 1.0 pu rapidly.

4 Exercise 3: Connection to an Infinite Bus

Generator A is replaced by an ideal three-phase voltage source representing an **infinite bus** — a bus that maintains constant terminal voltage and constant frequency regardless of the active or reactive power exchanged with it. This models a large national power grid to which a small generator is connected. By definition, the infinite bus has zero impedance ($Z_\infty \to 0$) and infinite apparent power capacity ($S_\infty \to \infty$).

The fundamental consequence is that the infinite bus completely dominates frequency: no action of Generator B can alter the system frequency. Generator B must adapt entirely to the grid's frequency, not the other way around.

4.1 Simulink Model

Figure 18 shows the Exercise 3 Simulink model. Generator A has been replaced by an ideal three-phase voltage source (infinite bus) that maintains constant voltage at 1.0 pu and constant frequency at 60 Hz. Generator B connects through the synchronisation breaker at $t = 5\,\text{s}$. The infinite bus absorbs or supplies any real and reactive power demanded by Gen B without any change in its terminal conditions.

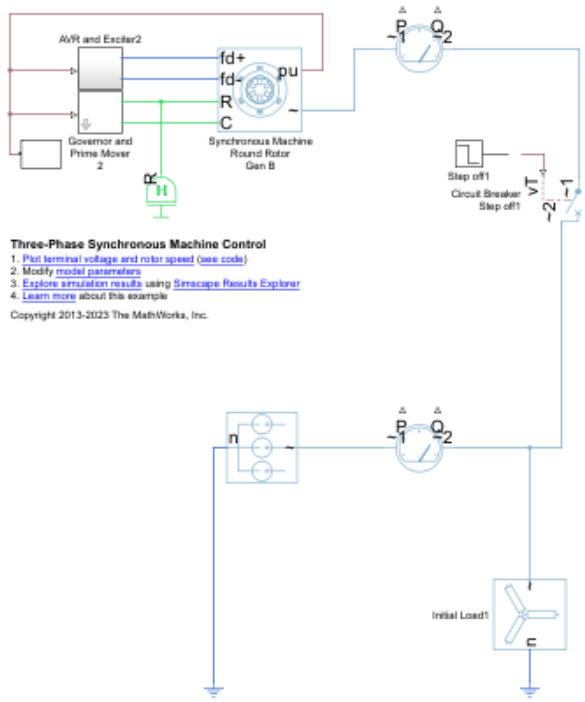

Figure 18. Exercise 3 Simulink model: Generator B connected to an infinite bus (ideal voltage source) through a synchronisation breaker. The infinite bus maintains $V = 1.0\,\text{pu}$ and $f = 60\,\text{Hz}$ at all times.

4.2 Task 5 – Generator B at Lower Frequency (0.98 pu) vs Infinite Bus

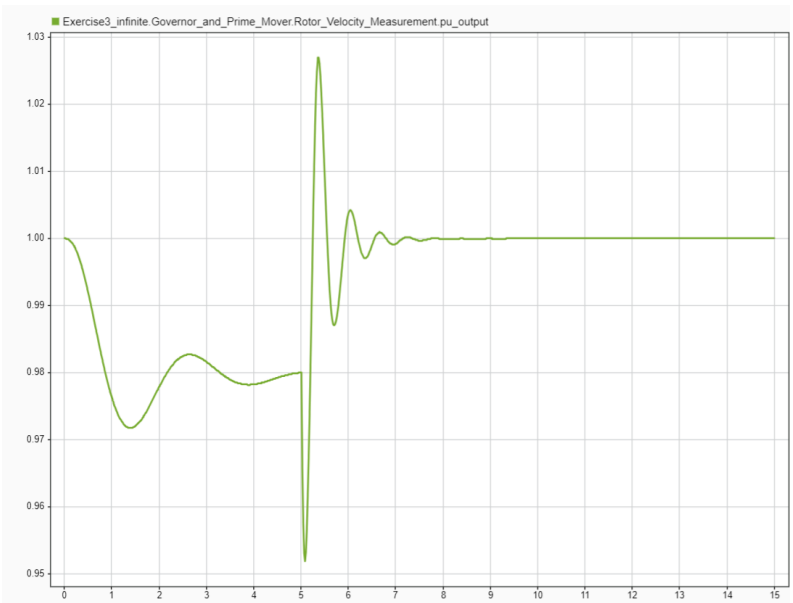

Figure 19. Task 5: Frequency variation when Generator B is paralleled with an infinite bus at $f_{\text{set}} = 0.98\,\text{pu}$. System frequency remains rigid at $1.000\,\text{pu}$.

Upon breaker closure at $t = 5\,\text{s}$, the infinite bus instantaneously forces Gen B's frequency to 1.0 pu. The system frequency does not change at all — it remains exactly 1.0 pu throughout. This is the defining characteristic of an infinite bus and the fundamental contrast with Task 3.

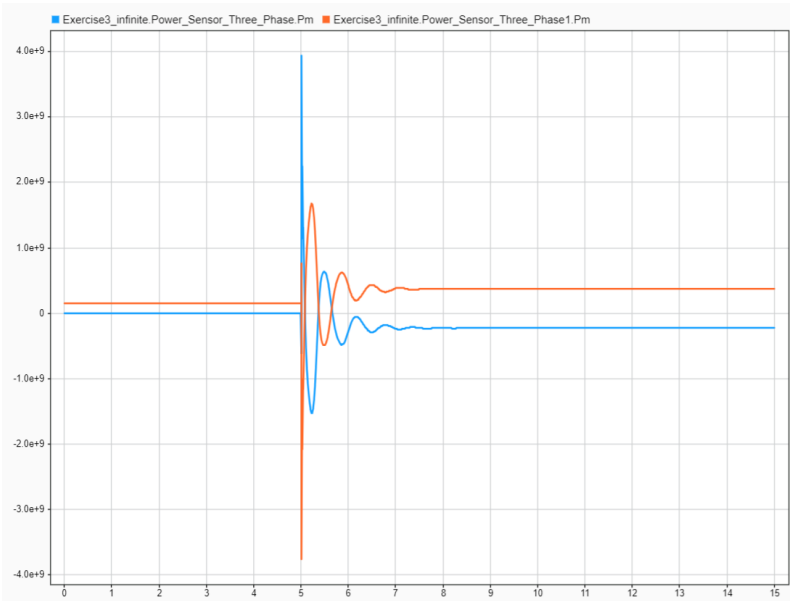

Figure 20. Task 5: Active power when Generator B is paralleled with an infinite bus at $f_{\text{set}} = 0.98\,\text{pu}$. Gen B absorbs power (severe motoring condition).

With the grid frequency fixed at 1.0 pu and Gen B's setpoint at 0.98 pu, the droop equation gives:

$$P_B = \frac{f_{\text{setpoint}} - f_{\text{grid}}}{S_P} = \frac{0.98 - 1.0}{S_P} < 0 \tag{4.1}$$

Gen B acts as a synchronous motor, absorbing power from the infinite bus. In Task 3, the system frequency settled at a compromise of about 1.008 pu, giving a deficit of $1.008 - 0.98 = 0.028\,\text{pu}$. Here, the infinite bus keeps the frequency at exactly 1.0 pu, so the deficit is only $1.000 - 0.98 = 0.020\,\text{pu}$ — numerically smaller. But the practical problem is still serious: Gen B cannot pull the system frequency down like it did in Task 3, so the motoring condition stays until the operator either raises the governor setpoint or opens the breaker.

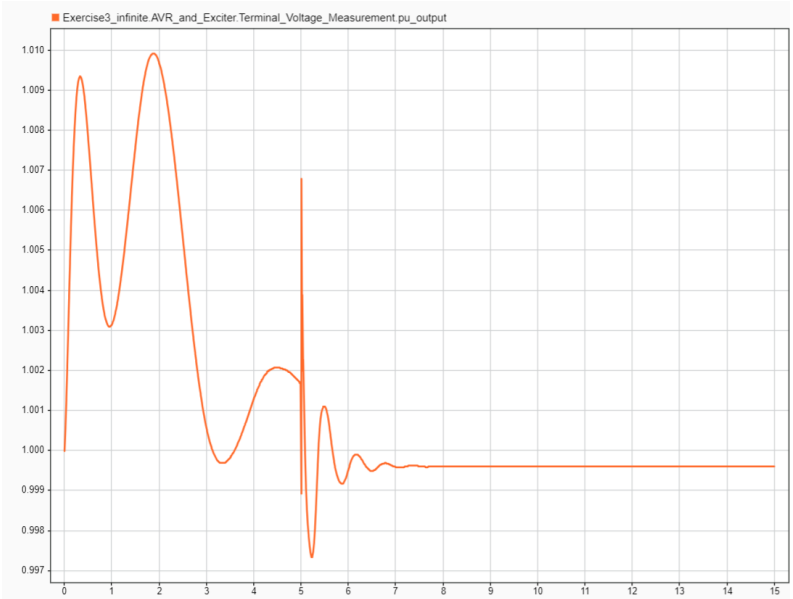

Figure 21. Task 5: Terminal voltage when Generator B is connected to an infinite bus at $f_{\text{set}} = 0.98\,\text{pu}$.

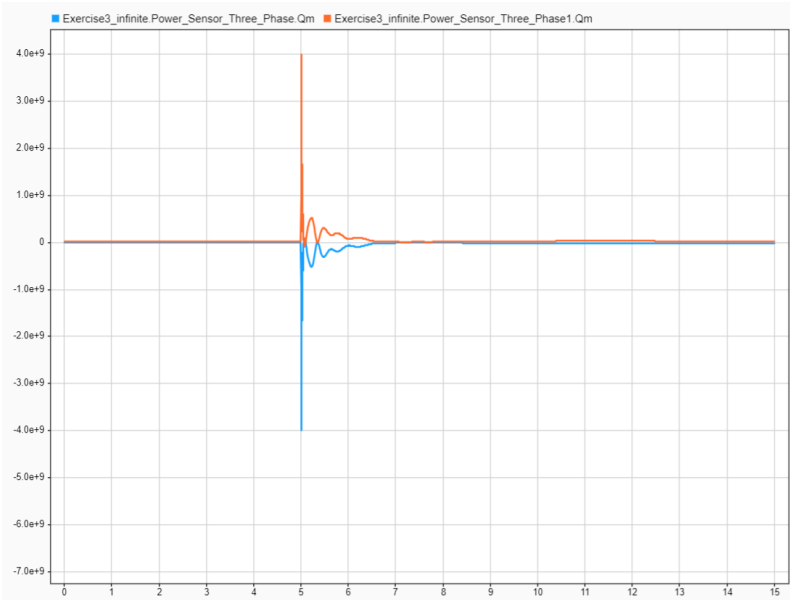

Figure 22. Task 5: Reactive power when Generator B is connected to an infinite bus at $f_{\text{set}} = 0.98\,\text{pu}$.

Upon breaker closure, the terminal voltage experiences a brief transient dip caused by the circulating current between Gen B and the infinite bus. Because the infinite bus holds its voltage at exactly 1.0 pu, Gen B's AVR is the only regulator that needs to act; the grid-side voltage is unaffected. The AVR increases field excitation to restore V_t, which also raises the reactive power output of Gen B. In steady state, Gen B settles at a reactive power level determined by the difference between its internal EMF and the bus voltage. Compared with Task 3, where both AVRs participated in voltage recovery, the infinite bus case places the entire reactive power burden on Gen B's excitation system.

4.2.1 Comparison with Task 3

Table 2 quantifies the contrast between the two scenarios.

Table 2. Task 3 vs Task 5 – Finite Bus vs Infinite Bus ($f_{\text{set,B}} = 0.98\,\text{pu}$)

Parameter	Task 3 (Finite Bus)	Task 5 (Infinite Bus)
System frequency after sync	$\approx$ 1.008 pu (compromise between setpoints)	1.000 pu exactly (rigid, unchanged)
Gen B active power	Slightly negative (weak motoring)	Strongly negative (severe motoring)
Freq. influence of Gen B	Drags system freq. down partially	Zero influence on system freq.
Voltage regulation	Both AVRs act to restore 1.0 pu	Grid holds voltage; AVR acts on Gen B only
Engineering implication	Overloads Gen A; unstable if repeated	Gen B trips on overload; must raise setpoint

4.3 Task 6 – Generator B at Higher Frequency (1.02 pu) vs Infinite Bus

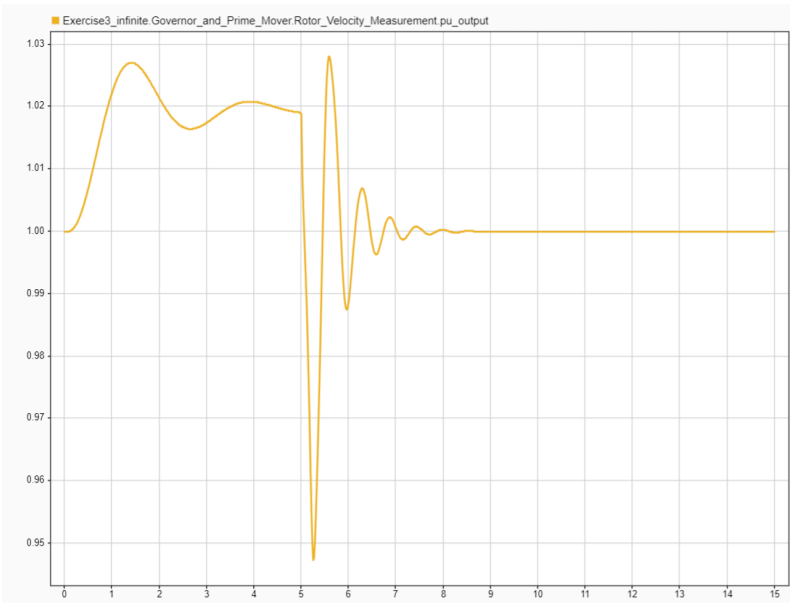

Figure 23. Task 6: Frequency variation when Generator B is paralleled with an infinite bus at $f_{\text{set}} = 1.02\,\text{pu}$. System frequency remains rigid at 1.000 pu.

Generator B is spun up to 1.02 pu before synchronisation. Upon breaker closure, the infinite bus immediately locks Gen B to 1.0 pu. The frequency does not rise — it remains exactly 1.0 pu. This is the key contrast with Task 4, where Gen A (a finite machine) was pulled upward in frequency by Gen B.

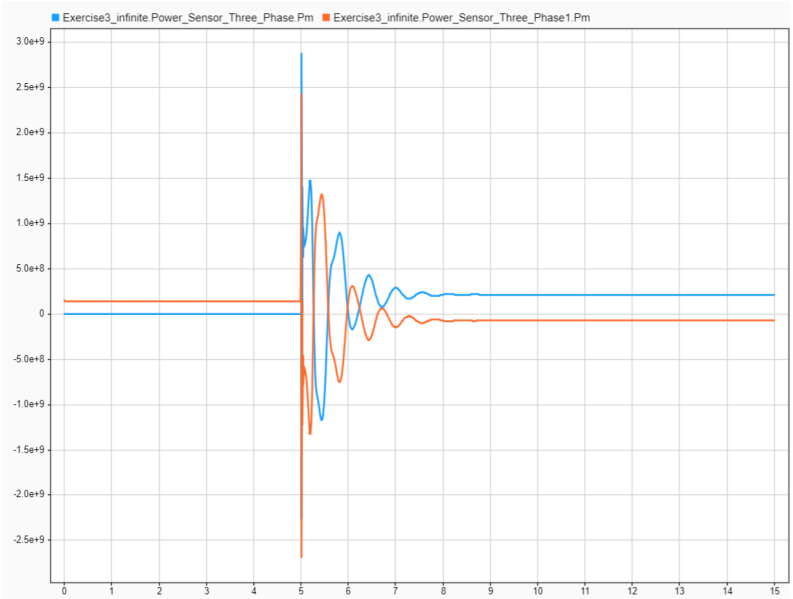

Figure 24. Task 6: Active power when Generator B is paralleled with an infinite bus at $f_{\text{set}} = 1.02\,\text{pu}$. Gen B injects power into the grid.

With Gen B's setpoint above the infinite bus frequency:

$$P_B = \frac{f_{\text{setpoint}} - f_{\text{grid}}}{S_P} = \frac{1.02 - 1.0}{S_P} > 0 \tag{4.2}$$

Generator B injects positive active power into the infinite bus. The grid absorbs this power effortlessly without any frequency perturbation.

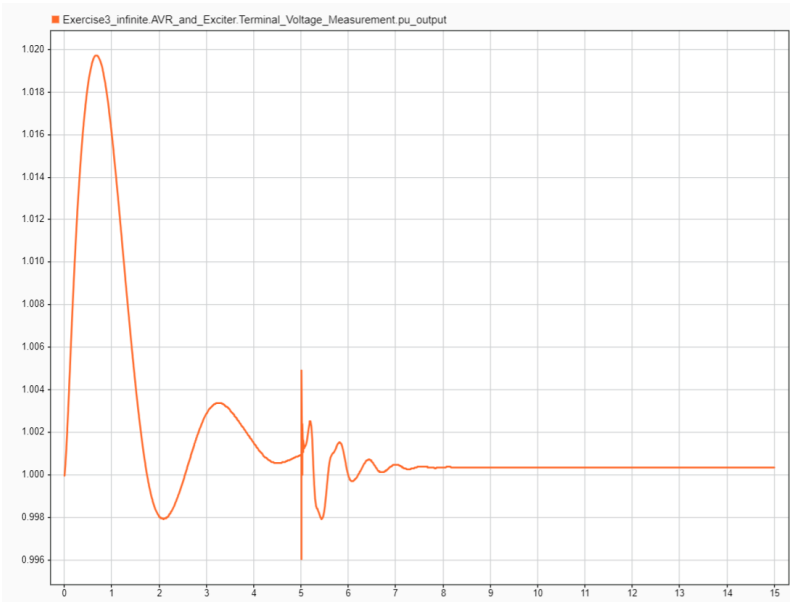

Figure 25. Task 6: Terminal voltage when Generator B is connected to an infinite bus at $f_{\text{set}} = 1.02\,\text{pu}$.

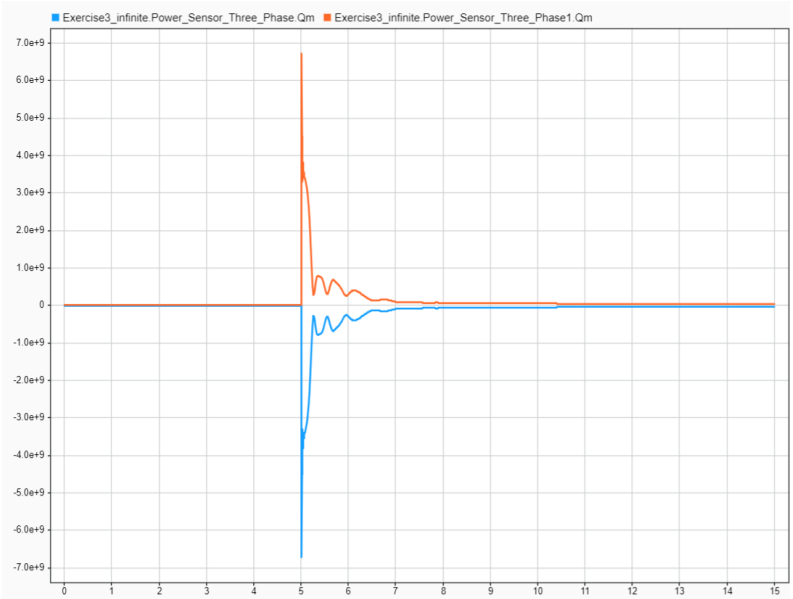

Figure 26. Task 6: Reactive power when Generator B is connected to an infinite bus at $f_{\text{set}} = 1.02\,\text{pu}$.

The voltage transient at breaker closure in Task 6 is a brief overshoot rather than a dip, because Gen B's rotor was leading the infinite bus in phase angle at the moment of connection. The AVR damps this overshoot within one to two seconds. In steady state, Gen B exports reactive power set by the difference between its internal EMF (driven by the AVR setpoint) and the fixed bus voltage of 1.0 pu. The infinite bus absorbs any surplus reactive power without any voltage deviation, which contrasts with Task 4 where both generators shared the reactive power adjustment. This clean voltage recovery confirms that the AVR–exciter loop operates independently of the P–f loop, consistent with the decoupled control design discussed in Exercise 1.

4.3.1 Comparison with Task 4

Table 3 quantifies the contrast.

Table 3. Task 4 vs Task 6 – Finite Bus vs Infinite Bus ($f_{\text{set,B}}$: 1.02 pu)

Parameter	Task 4 (Finite Bus)	Task 6 (Infinite Bus)
System frequency after sync	Raised by Gen B (droop-weighted average)	1.000 pu exactly (cannot be raised)
Gen A / Grid response	Gen A reduces output significantly	Grid absorbs all Gen B output silently
Gen B active power	Majority of load (generation)	Fixed by setpoint offset from grid
Voltage behaviour	Spike then 1.0 pu	Spike then 1.0 pu (identical mechanism)
Freq. influence of Gen B	Raises system frequency	Cannot raise grid frequency
Engineering implication	Load transferred; useful commissioning	Normal grid-tie generation; power sold

4.4 Comparison – All Tasks

The following comparison across all four parallel-operation tasks (Tasks 3–6) shows how the bus type (finite vs infinite) and the governor setpoint (under-frequency vs over-frequency) interact to determine system behaviour.

Key observations are summarised in Table 4.

Table 4. Comparison of Tasks 3–6

Parameter	Task 3 (Finite, 0.98)	Task 4 (Finite, 1.02)	Task 5 (Infinite, 0.98)	Task 6 (Infinite, 1.02)
System freq. after sync	≈ 1.008 pu (compromise)	≈ 1.015 pu (compromise)	1.000 pu (rigid)	1.000 pu (rigid)
Gen B active power	Negative (motoring)	Positive (generation)	Negative (motoring)	Positive (generation)
Gen B freq. influence	Pulls system freq. down	Pulls system freq. up	None	None
Voltage regulation	Both AVRs act	Both AVRs act	Grid holds V; Gen B AVR only	Grid holds V; Gen B AVR only
Engineering implication	Overloads Gen A	Load transferred successfully	Gen B trips on overload	Normal grid-tie operation

The governor setpoint relative to the system frequency determines whether a generator motors or generates in every scenario. The bus type (finite vs infinite) controls whether the oncoming machine can influence system frequency: a finite bus allows a droop-weighted compromise, while an infinite bus holds frequency immovable at 1.0 pu.

5 Conclusion

This report looked at synchronous generator behaviour under three increasingly complex scenarios using MATLAB R2025b / Simscape Electrical. The results were checked against standard power systems equations.

Exercise 1 covered the two main control loops of a synchronous machine: the swing-equation/governor loop linking active power to frequency, and the Q-equation/AVR loop linking reactive power to terminal voltage. The 555 MVA machine recovered from a 0.82 pu voltage dip within 3–4 seconds thanks to AVR action, and the frequency settled at values matching the droop prediction from (2.9).

Exercise 2 showed that the governor frequency setpoint controls load sharing between parallel generators. When the oncoming machine has a lower setpoint (Task 3), it draws power from the system, which is a dangerous motoring condition. A higher setpoint (Task 4) makes the new machine take over most of the load, which is the intended outcome. In both cases, the system frequency settled at a droop-weighted average of the two setpoints, consistent with (3.2).

Exercise 3 showed how an infinite bus differs from a finite one. Connected to an infinite bus, Generator B has no effect on system frequency. Its power output is set entirely by its governor setpoint relative to the fixed grid frequency — Task 5 resulted in motoring and Task 6 gave a clean power injection. In practice, this is what happens when a small generator connects to a national grid. The grid frequency is set by the combined inertia of millions of megawatts, and one machine cannot change it.

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[3] IEEE Std 421.5-2016, "IEEE Recommended Practice for Excitation System Models for Power System Stability Studies," IEEE, 2016.

[4] IEEE Std 1110-2019, "IEEE Guide for Synchronous Generator Modeling Practices and Parameter Verification with Applications in Power System Stability Analyses," IEEE, 2019.